

[Packages, Enum class, Wrapper class]

Program-1:

```
J Emp.java > ...
1  public class Emp
2  {
3      enum Gender
4      {
5          MALE,
6          FEMALE
7      }
8
9      Run | Debug | Run main | Debug main
10     public static void main(String[] args)
11     {
12         int eid=101;
13         String name="ABC";
14
15
16         System.out.println("eid="+eid);
17         System.out.println("ename="+name);
18
19         Gender gender_select=Gender.MALE;
20         System.out.println("gender="+gender_select);
21     }
22 }
23
```

Program-2:

```
1  enum City
2  {
3      BARODA,
4      AHMEDABAD,
5      SURAT
6  }
7  public class City_detail
8  {
9      Run | Debug | Run main | Debug main
10     public static void main(String[] args)
11     {
12         City c1=City.BARODA;
13         City c2=City.SURAT;
14
15         System.out.println("first city="+c1);
16         System.out.println("Second city="+c2);
17     }
18 }
19
```

Program-3:

```
1  public class Enum_methods
2  {
3
4      public enum Direction
5      {
6          NORTH, SOUTH, EAST, WEST
7      }
8
9      Run | Debug | Run main | Debug main
10     public static void main(String[] args) {
11         // Using values() method it return all values mention in given enum areas
12         for (Direction dir : Direction.values())
13         {
14             System.out.println(dir);
15
16             // Using valueOf() method, it will display specified values in string format
17             Direction dir_value = Direction.valueOf("EAST");
18             System.out.println("Direction: " + dir_value);
19
20             // Using ordinal() method, it will identify the position of specific values
21             System.out.println("Ordinal of EAST: " + dir_value.ordinal());
22         }
23     }
24
25
```

Program:4

```
1  class enum_constructor {
2
3
4      enum Enums
5      {
6          A, B, C;
7
8          {
9              System.out.println(x: 1);
10         }
11
12         static
13         {
14             System.out.println(x: 2);
15         }
16
17         private Enums()
18         {
19             System.out.println(x: 3);
20         }
21     }
22
23     Run | Debug | Run main | Debug main
24     public static void main(String[] args)
25     {
26         Enums en = Enums.C;
27     }
28 }
```

Program-5:

```
J simple_pack.java > ...
1  package abc;
2
3  public class simple_pack {
    Run main | Debug main | Run | Debug
4      public static void main(String args[]) {
5          System.out.println(x: "Welcome to package");
6      }
7  }
8
9
10
11
12  /*
13      * set classpath="g:\java_prog\packages" ;
14      * javac -d . simple_pack.java
15      * java simple_pack.simple_pack
16      */
```

Program-6:

First file:

```
1  package student;
2  public class s1
3  {
4      public void sdata()
5      {
6          String sname="a1";
7          int rollno=1;
8          int age=17;
9          System.out.println("student name="+sname);
10         System.out.println("student rollno="+rollno);
11         System.out.println("student age="+age);
12     }
13 }
14 }
```

Second file:

```
1
2  import student.s1;
3
4  public class s2
5  {
6      Run main | Debug main | Run | Debug
7      public static void main(String args[])
8      {
9          s1 obj = new s1();
10         obj.sdata();
11     }
```

Program:7

First file:

```
J A.java > Language Support for Java(TM) by Red Hat > { } mypack1
1  package mypack1;
2  public class A
3  {
4      public void msg()
5      {
6          System.out.println(x: "Hello");
7      }
8  }
```


Second File:

```
J B.java > ...  
1  import mypack1.A;  
2    
3  //package mypack1;  
4  class B {  
    Run main | Debug main | Run | Debug  
5  public static void main(String[] args) {  
6      A obj = new A();  
7      obj.msg();  
8  }  
9  }
```

Program-8:

```
J First.java > Language Support for Java(TM) by Red Hat > First > main(String[])
1  public class First {
    Run main | Debug main | Run | Debug
2      public static void main(String[] args)
3      {
4          //create autoboxing processs
5
6          // create primitive types
7          int a = 5;
8          double b = 5.65;
9
10
11         //converts into wrapper objects
12         Integer obj=new Integer(value: 50);
13
14
15         Integer a1 = Integer.valueOf(a);
16         Double b1 = Double.valueOf(b);
17
18         System.out.println(a1.TYPE);
19         System.out.println(a1.SIZE);
20
21         if(a1 instanceof Integer) {
22             System.out.println(x: "An object of Integer is created.");
23         }
24
25         if(b1 instanceof Double) {
26             System.out.println(x: "An object of Double is created.");
27         }
28     }
```

Program-9:

```
Second.java > ...
1  public class Second {
2
3      Run main | Debug main | Run | Debug
4      public static void main(String[] args)
5      {
6          // Float f1=new Float();
7          String s="3.14f";
8          System.out.println(s);
9          Float f1=Float.valueOf(s);
10         System.out.println("float="+f1);
11
12         Integer i=Integer.valueOf(s: "101011",radix: 2);
13         System.out.println(i);
14
15
16         //parseXXX()
17         int i1=Integer.parseInt(s: "12345");
18         System.out.println(i1);
19
20         float f=Float.parseFloat(s: "3.14f");
21         System.out.println(f);
22
23         //xxxValue()
24         Float f2=Float.parseFloat(s: "3.14f");
25         float fvalue=f2.floatValue();
26         System.out.println("xvalue()"+fvalue);
27     }
28 }
```

Program-10:

```
J Type_Conversation.java > Language Support for Java(TM) by Red Hat > Type_Conversation > main(String[])
1  public class Type_Conversation
2  {
   Run main | Debug main | Run | Debug
3  public static void main(String[] args) {
4      String a=new String(original: "10");
5
6      String b=new String(original: "20");
7
8      System.out.println(a+b);
9
10
11     int sum=Integer.parseInt(a)+Integer.parseInt(b);
12
13     System.out.println("sum of given value="+sum);
14
15
16 }
17
18
```

1.	Write a Java program to create an enum called "DaysOfWeek" representing the days of the week.
2.	Write a Java program to implement an enum called "Shape" with constants representing different geometric shapes.
3.	Write a java program to swap two numbers using by third variable by defining UDF function and execute this swapping process package.
4.	Write a java program to check inputted number is a even or odd which execute by creating a package name "EvenOdd".